

INGALLS FIELD, VA, USA

WMO: 724115

Lat: 37.950N

Lon: 79.833W

Elev: 1156

StdP: 88.19

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 93757

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.2	-13.9	-22.9	0.5	-11.9	-20.2	0.7	-9.0	15.0	-5.9	14.1	-4.3	8.4	290	0.557

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.4	27.7	19.9	26.4	19.4	25.1	18.8	22.3	25.0	21.3	24.1	20.6	23.4	3.6	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.4	18.5	23.5	20.9	17.9	22.8	19.7	16.7	21.8	71.6	24.9	68.1	24.0	64.9	23.8	26.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.5	11.4	10.3	-20.2	29.6	3.9	2.4	-23.0	31.4	-25.3	32.8	-27.5	34.2	-30.3	35.9		
(5)			-20.1	23.0	3.5	1.5	-22.6	24.1	-24.6	25.0	-26.6	25.9	-29.1	27.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	9.8	-2.0	-0.3	3.6	9.6	14.4	18.5	20.6	16.5	10.9	4.7	0.6	(6)	
(7)		DBStd	9.53	7.16	6.59	6.42	5.75	4.96	3.37	2.69	3.06	3.89	5.01	5.89	6.30	(7)
(8)		HDD10.0	1503	377	291	215	77	18	1	0	1	3	50	174	296	(8)
(9)		HDD18.3	3343	632	521	457	266	135	37	11	17	78	232	408	549	(9)
(10)		CDD10.0	1437	3	3	17	64	154	257	329	312	198	79	16	4	(10)
(11)		CDD18.3	236	0	0	0	3	13	42	81	69	24	3	0	0	(11)
(12)		CDH23.3	777	0	0	1	12	46	122	293	224	71	9	0	0	(12)
(13)		CDH26.7	100	0	0	0	1	5	9	45	29	7	4	0	0	(13)
(14)	Wind	WSAvg	4.8	6.5	6.1	5.4	5.2	4.4	4.0	3.6	3.2	3.5	4.6	5.3	5.8	(14)
(15)	Precipitation	PrecAvg	1007	81	66	96	82	99	97	100	77	93	60	76	79	(15)
(16)		PrecMax	1427	177	162	180	134	198	221	158	119	249	170	152	169	(16)
(17)		PrecMin	843	25	17	30	25	21	24	42	24	9	0	11	41	(17)
(18)		PrecStd	141	43	35	39	31	46	51	34	27	63	45	38	30	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.0	16.3	21.0	25.1	27.1	27.8	29.8	27.7	23.9	18.9	16.0	(19)	
(20)			MCWB	12.4	11.9	12.7	14.5	17.0	19.8	20.0	20.9	18.5	18.1	13.4	13.2	(20)
(21)		2%	DB	12.7	13.2	17.5	22.3	24.2	26.3	27.7	27.4	25.6	21.4	16.5	13.0	(21)
(22)			MCWB	10.7	10.4	11.2	13.4	16.8	19.5	20.1	20.6	18.2	15.9	11.2	11.0	(22)
(23)		5%	DB	11.0	11.2	15.0	19.9	22.7	24.8	26.3	26.0	23.4	19.1	14.1	11.9	(23)
(24)			MCWB	9.5	8.1	9.6	12.2	16.4	18.8	19.7	20.0	17.5	14.9	10.3	9.7	(24)
(25)		10%	DB	7.9	8.9	12.6	17.6	21.2	23.0	24.9	24.0	22.1	17.6	12.7	9.7	(25)
(26)			MCWB	6.0	6.2	8.4	11.5	15.8	18.3	19.2	19.1	17.0	13.9	9.5	7.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.2	13.7	14.4	17.0	20.1	22.2	23.6	23.2	21.8	20.1	15.1	14.2	(27)
(28)			MCD B	14.6	15.3	18.5	21.2	23.0	24.9	26.3	26.4	23.9	22.5	17.3	14.9	(28)
(29)		2%	WB	11.4	11.4	12.7	15.4	18.7	20.9	22.4	22.2	20.3	17.9	13.4	12.2	(29)
(30)			MCD B	12.3	13.0	15.9	19.1	22.1	24.0	24.9	25.1	22.7	19.5	15.1	13.0	(30)
(31)		5%	WB	9.3	8.9	11.2	14.0	17.7	20.1	21.3	21.2	19.2	16.2	12.1	10.2	(31)
(32)			MCD B	10.3	10.4	13.9	17.4	20.9	23.3	24.0	24.2	21.5	18.4	13.5	11.3	(32)
(33)		10%	WB	6.7	6.6	9.3	12.8	16.7	19.3	20.5	20.4	18.2	14.6	10.3	7.9	(33)
(34)			MCD B	7.9	8.4	11.8	16.2	19.8	22.4	23.3	23.2	20.3	17.0	12.3	8.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.8	8.3	8.9	9.4	8.8	7.8	7.4	7.5	7.4	7.7	7.7	7.2	(35)
(36)			MCDBR	7.7	10.0	10.1	10.7	9.9	8.7	8.4	8.4	8.6	8.8	8.5	7.6	(36)
(37)			MCWBR	6.5	8.0	6.4	5.5	5.3	4.8	4.3	4.2	4.4	5.3	6.3	6.4	(37)
(38)		5% WB	MCD BR	7.8	9.7	9.3	9.0	8.8	7.7	7.3	7.5	7.2	7.7	7.8	7.8	(38)
(39)			MCWBR	7.3	9.0	6.7	5.9	5.3	4.5	4.4	4.1	4.2	5.5	7.0	7.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.270	0.284	0.307	0.342	0.373	0.402	0.423	0.425	0.342	0.314	0.284	0.269	(40)	
(41)		taud	2.545	2.491	2.484	2.408	2.314	2.258	2.230	2.195	2.436	2.515	2.574	2.574	(41)	
(42)		Ebn at Noon	931	955	959	936	907	877	855	844	908	906	906	909	(42)	
(43)		Edh at Noon	74	90	100	115	129	136	139	140	104	87	72	68	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.10	2.85	3.90	4.94	5.44	5.97	5.68	5.19	4.34	3.36	2.46	1.84	(44)	
(45)		RadStd	0.19	0.36	0.25	0.39	0.47	0.40	0.35	0.26	0.47	0.40	0.23	0.16	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	-1.04	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (33 neighbors)		+0.35	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+126	+64

Nomenclature: See separate page